Sparse Dictionary Learning for Interpretable Lung Disease Diagnosis from 3D Chest CT

An undergraduate project proposal report submitted to the

Department of Electrical and Information Engineering
Faculty of Engineering
University of Ruhuna
Sri Lanka

in partial fulfillment of the requirements for the

Degree of the Bachelor of the Science of Engineering Honours

by

K.R.C.S. Kekulawala	-	EG/2021/4609
M.D.Y.B. Kularathne	-	EG/2021/4620
B.G.P.U. Thilakshana	-	EG/2021/4831
B.P.G.L. Thejana	-	EG/2021/4828

....
Dr. Kaveen Liyanage
(Supervisor)

....
Mrs. Sithara Mahagama
(Co-Supervisor)

Abstract

Abstract goes here.

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Acronyms

AI Artificial Intelligence

C/G Consolidation-to-Ground-Glass-Opacity

CNN Convolutional Neural Network

CT Computed Tomography

GGO Ground-Glass Opacity

HU Hounsfield Unit

KSVD K-Singular Value Decomposition

LC-KSVD Label-Consistent K-Singular Value Decomposition

Chapter 1

Introduction

1.1 Background

Lung disease remains a leading cause of global morbidity and mortality. In the diagnosis of lung disease, assessing the pattern, location, and regional distribution of involvement is the domain of radiology. Clinical diagnosis increasingly relies on radiological patterns to differentiate between pathologies with overlapping symptoms. While Chest X-rays (CXR) remain the first-line screening tool due to low cost and radiation, they suffer from limited sensitivity (roughly 70% visibility of lung volume) due to anatomical superposition, (where all anatomical structures located in the path of the X-rays are stacked onto a 2D plane). [1]

Computed Tomography (CT) is a radiographic technology that combines X-ray equipment with a computer and a cathode ray tube display to produce images of cross sections of the human body. [2] overcomes many of the limitations of projection-based radiography by eliminating anatomical superposition and enabling full volumetric assessment of the lungs. By combining X-ray acquisition with computational reconstruction, CT produces high-resolution slices that reveal fine structural details not visible in standard chest radiographs

A fundamental metric of a CT scan is the Hounsfield Unit (HU), a relative quantitative scale that describes radiodensity [1.1] In pulmonary imaging, HU values allow for the objective differentiation of tissues such as uch as normal parenchyma, Ground-Glass Opacity (GGO), and consolidations that might appear visually similar.

GGO appears as hazy increased opacity of lung, with preservation of bronchial and vascular margins (Fig 1). Consolidation on the other hand, appears as a homogeneous increase in pulmonary parenchymal attenuation that obscures the margins of vessels and airway walls (45) (Fig 2). Consolidation is more opaque than GGO. While GGO often suggests a reversible or early-stage process, consolidation usually indicates a more aggressive or advanced stage. The Consolidation-to-GGO (C/G) ratio is a quantitative metric calculated by dividing the volume or area of consolidation by the volume or area of ground-glass opacity within a lesion or throughout the lungs.

The traditional approach for CT analysis relies on a ”slice-by-slice” approach where a radiologist views a series of axial cross-sections and identifies the ”index slice” (the single image where a pathology appears at its largest diameter). A 2D slice only captures a small fraction of a 3D structure. If a nodule is irregularly shaped, the "largest" slice may not represent the true extent of the disease [2.2]. Studies show that 2D measurements are highly sensitive to the patient's orientation and the specific slice chosen, leading to high inter-observer variability [2.3]. 2D models treat each slice as an independent data point, losing the "spatial continuity" between adjacent tissues [1.4].

3D volumetric analysis represents a quantitative shift wherein the entire dataset is used to reconstruct the lung's anatomy as a solid volume. This enables more accurate characterization of lesion size, shape, and spatial distribution, while reducing sensitivity to slice selection and viewing angle [5.1]. By preserving inter-slice context, 3D representations better capture disease morphology and progression, providing a more robust foundation for quantitative analysis and automated learning-based methods.

Modern radiology increasingly relies on Artificial Intelligence (AI) to streamline the acquisition of radiologist analyses on chest X-rays, reinforcing the potency of automated triaging systems in streamlining healthcare workflows and amplifying patient care standards [42][1]. However, the exponential increase in data volume generated by 3D CT scanners presents a profound challenge for human interpretation and computational analysis. While the prevailing research trend has focused heavily on the deployment of 3D deep learning architectures, such as Convolutional Neural Networks (CNNs), these models frequently encounter significant hurdles regarding computational overhead, data scarcity, and a critical lack of interpretability. The research proposed herein advocates for a departure from these "black box" paradigms, suggesting that sparse representation methods offer a superior framework for the volumetric identification of lung diseases by ensuring clinical interpretability.

1.2 Problem Statement

Modern radiology increasingly relies on AI to manage the vast quantities of 3D data produced during routine screenings. 3D deep learning has been widely applied to the automatic diagnosis of lung diseases, achieving remarkable success in tasks such as segmenting suspected infection regions and differentiating between various types of carcinoma. Despite these advancements, the inherent complexity of 3D CNNs introduces a set of constraints that are often overlooked in the pursuit of high accuracy scores.

1.2.1 Computational Overhead

Processing a 3D CT scan involves treating stacked 2D slices as a single volumetric matrix, which demands significantly more GPU memory and processing power than standard image analysis. Unlike ordinary RGB images, CT scans consist of single-channel data with varying voxel dimensions and resolution, which complicates spatial alignment and generalizability across different scanners. Furthermore, the "curse of dimensionality" remains a persistent threat; as the network depth increases to capture global features, the risk of gradient explosion and slow convergence becomes more pronounced.

1.2.2 Interpretability Bottleneck

Beyond computational limits, the clinical utility of deep learning is hampered by the interpretability paradox. While tools like Grad-CAM provide visual heatmaps to localize lesions, these post-hoc explanations are often insufficient for medical professionals who require a granular understanding of the model's decision-making process. In high-stakes environments, a model that shruggingly produces a diagnosis without an audit trail of its reasoning is difficult to integrate into standard workflows, as it lacks the accountability necessary for legal and ethical compliance. This transparency gap has led to a renewed interest in "open book" models that are interpretable by design, where the internal logic is clear from the outset.

1.3 Objectives and Scope

1.3.1 Objectives

1. Develop a Dictionary Learning framework for Interpretable Lung Disease Classification from 3D Chest CT
   - Implement training in two phases for normal and pathological atoms
   - Integrate KSVD with label consistency constraints
2. Visualize dictionary atoms and map them to anatomical features
   - Compare sparse coefficient patterns across disease classes
   - Visulaize dictionary atoms using an interactive 3D interface
   - Demonstrate atom-to-pathology correspondence
3. Benchmark against baseline models
   - Compare with standard KSVD without label consistency constraint
   - Compare with 3D CNN baseline (3D-ResNet)
4. Integrate clinical metrics for volumetric analysis
   - Implement HU-based segmentation for GGO and consolidation
   - Calculate C/G ratio and compare model predictions against clinical thresholds

1.3.2 Scope

- The proposed solution is limited to volumetric CT scans of the thorax and does not consider other imaging modalities such as chest X-rays, MRI, or PET scans.
- The methodology is restricted to dictionary learning–based approaches, with particular emphasis on KSVD and LC-KSVD End-to-end deep dearning models such as CNNs are not the primary focus of this work and are included only as baseline comparators for performance benchmarking.

- Clinically, the analysis is confined to the characterization of lung parenchymal abnormalities, specifically GGO and consolidation patterns. Quantitative metrics such as HU-based segmentation and the C/G ratio are employed to support volumetric analysis. The study does not address disease prognosis or treatment outcome prediction.

- The scope excludes real-time clinical deployment, cross-institutional generalization studies, and regulatory validation. The findings are intended to demonstrate the feasibility and interpretability of sparse representation learning for 3D lung disease analysis rather than to provide a clinically deployable diagnostic system.

Chapter 2

Literature Review

2.1 Previous Work

2.1.1 Deep Learning for Lung Disease Diagnosis from Medical Imaging

The field of automated lung disease diagnosis has seen a proliferation of methodologies, each attempting to balance accuracy with computational feasibility. Early approaches relied on handcrafted features and classical classifiers like Support Vector Machines (SVM) and Random Forests. While these methods provided strong interpretability, they often lacked the flexibility to handle the high variability of CT data. The emergence of deep learning fundamentally changed this landscape.especially in areas such as;

- Image Segmentation and Classification
- Advancing Diagnostics with AI and CAD Systems
- Prognostics with Radiomics and Predictive Analytics
- Workflow Optimisation Using AI

Convolutional Neural Networks (CNNs) have emerged as the dominant paradigm for automated analysis of chest computed tomography (CT) scans. Recent surveys indicate that deep learning models achieve classification accuracy exceeding 98% for multi-disease detection tasks, representing a marked improvement over traditional machine learning approaches. Contemporary research has increasingly focused on 3D CNN architectures that treat CT scans as complete volumetric data rather than independent 2D slices. This volumetric approach captures spatial relationships within lung tissue more effectively, leading to improved sensitivity in detecting subtle pathological changes. Li et al. (2025) demonstrated that preprocessing techniques such as lung region extraction, combined with advanced backbone architectures like ResNeSt50, significantly enhance classification performance for multiple lung pathologies [3][4].

2.1.2 Sparse Dictionary Learning in Medical Image Analysis

Sparse dictionary learning represents an unsupervised machine learning paradigm for extracting interpretable, meaningful features from medical images. This technique learns a set of dictionary elements or basis vectors that represent data in a sparse manner. In the context of 3D CT, a volumetric patch $y \in \mathbb{R}$ can be approximated by $y \sim Dx$, where $D \in \mathbb{R}^{\ltimes \frown \Bbbk}$ is the dictionary and $x \in \mathbb{R}^{\urcorner}$ is a sparse vector containing only a small number of non-zero coefficients.

The strength of dictionary learning lies in its ability to extract local, structural information from noisy medical images while requiring less labeled data than purely supervised deep learning approaches. This characteristic is particularly valuable for rare diseases with limited training samples.

The success of these models hinges on the quality of the learned dictionary. Traditional methods like K-SVD focus primarily on minimizing the reconstruction error, which is highly effective for image denoising and super-resolution but lacks the discriminative power required for classification. To address this, supervised dictionary learning techniques have emerged, which incorporate label information directly into the dictionary construction process.

The LC-KSVD algorithm enhances the traditional K-SVD by adding a label consistency constraint and a classification error term to the objective function. This approach forces features from the same class to have similar sparse codes, thereby increasing the discriminative power of the model.

2.1.3 Explainable AI in Medical Imaging

Clinical radiology requires an understandable method for determining how the disease occurs.The "black-box" nature of deep neural networks creates significant barriers to clinical adaptation. Physicians need to understand diagnostic rationale for accountability, patient safety, and effective care delivery. [9][10].

Predominant XAI approaches in medical imaging include:

- Visual explanation techniques like Integrated Gradients and Gradient-weighted Class Activation Mapping (Grad-CAM) produce saliency maps that highlight the areas of the image that have the greatest impact on model predictions [11].
- LIME (Local Interpretable Model-agnostic Explanations) and SHAP (SHapley Additive explanations) are two feature attribution methods that offer quantitative insight into feature contributions [12].
- Attention mechanisms that provide more comprehensible decision logic by allowing models to selectively focus on pertinent image regions [13].

According to studies, these explainability techniques greatly increase clinician comprehension and confidence in AI-driven diagnostic results, making it easier to integrate with conventional radiological workflows while maintaining regulatory compliance [14].

2.1.4 Volumetric Feature Extraction

The core of pulmonary diagnostic analysis lies in the identification and quantification of specific radiographic patterns. GGO is a hallmark sign in chest CT, characterized by hazy increased attenuation that does not obscure the underlying vascular markings. It is a non-specific finding that can represent fluid, fibrosis, or partial collapse of alveoli, and it is particularly prevalent in the early stages of diseases like COVID-19 or certain types of non-small cell lung cancer.

Volumetric analysis allows for the quantification of these features beyond simple 2D identification. By segmenting the lung parenchyma and applying Hounsfield Unit (HU) thresholding, researchers can calculate the "total pneumonia volume" or the percentage of lung tissue affected by specific patterns. A A critical metric in this domain is the Consolidation-to-GGO (C/G) ratio. Clinical data shows that as GGO progresses to consolidation, the severity of the patient's condition increases.

2.2 Gaps in Literature

2.2.1 Integration of interpretability with 3D volumetric analysis is limited.

While 3D CNNs and volumetric analysis methods have demonstrated superior performance for lung disease classification from CT scans, the majority of explainability research has focused on 2D imaging modalities (chest X-rays) or individual CT slices analyzed independently. There remains a significant gap in developing inherently interpretable feature representations specifically designed for full 3D volumetric data.

Existing post-hoc explanation methods for 3D data (e.g., 3D Grad-CAM) provide visualization of important regions but do not offer semantically meaningful feature representations that clinicians can directly relate to anatomical structures or pathological patterns. The extension of 3D saliency maps to interpretable concepts—such as specific lesion types, textural patterns, or anatomical landmarks—remains largely unexplored. This gap is particularly significant given that pulmonary pathology often manifests through spatial patterns spanning multiple CT slices (e.g., centrilobular versus panlobular emphysema distribution).

2.2.2 Lack of Inherently Interpretable Deep Learning Architectures

Modern XAI methodologies for medical imaging frequently utilize post hoc explanations for opaque deep learning models. These approaches explain actions taken afterward but fail to tackle the black-box issue—underlying models remain intricate and opaque, regardless of accompanying explanations.

Sparse dictionary learning presents an alternative approach where learned dictionary elements can represent interpretable features such as nodules, GGO, consolidations, and honeycombing. Every classification decision links to particular, tangible dictionary components, offering inherent rather than post-implementation clarity. Despite the integration of sparse dictionary learning principles with contemporary 3D deep learning models for analyzing lung CT scans, significant exploration in this area is still lacking. The existing hybrid methods, such as deep dictionary learning, have mainly been applied to 2D natural images rather than to volumetric medical imaging.

Frozen Dictionary Learning as a Mechanism for Pathological Differentiation

A primary innovation of the proposed system is the utilization of "frozen" dictionary learning. In conventional dictionary learning, all atoms are updated simultaneously, which can lead to a phenomenon where "normal" atoms inadvertently absorb pathological features, reducing the model's ability to clearly isolate anomalies. Frozen dictionary learning addresses this by employing a staged training strategy that partitions the dictionary into anatomically standard and pathologically specific components.

The first phase involves learning a base dictionary using only "normal" or healthy lung data. These atoms capture the expected appearance of lung parenchyma, bronchi, and pulmonary vasculature. Once this base dictionary is optimized, it is "frozen"-its atoms are held constant and are no longer updated. In the second phase, additional dictionary atoms are introduced and trained specifically on "anomalous" data, such as CT scans containing viral pneumonia or adenocarcinoma. Because the standard anatomical features are already accounted for by the frozen atoms, the new atoms are forced to capture only the unique variations or lesions that represent the disease state.

This method provides an unprecedented level of interpretability. When a query 3D patch is analyzed, the resulting sparse coefficient vector reveals exactly which atoms are being utilized. If the representation relies heavily on the "anomalous" portion of the dictionary, the clinician can identify the specific type of pathology present by mapping those atoms back to their visual counterparts in the original signal domain. This "feature separation" ensures that the diagnostic process is transparent and verifiable, moving away from the hidden logic of deep neural networks.

2.2.3 Handling Class Imbalance with Interpretable Models

Rare lung pathologies—certain cancer subtypes (e.g., large cell carcinoma, carcinoid tumors), uncommon ILD patterns (e.g., pleuroparenchymal fibroelastosis), and emerging infectious diseases—present severe class imbalance challenges. Deep learning research has addressed this through weighted loss functions, oversampling strategies, synthetic data generation (e.g., progressive growing GANs), and meta-learning approaches.

However, the application of class imbalance mitigation techniques within interpretable sparse representation frameworks for 3D CT analysis is underexplored. Dictionary learning methods that can effectively learn representative atoms for rare categories while maintaining interpretability for common pathologies require further development. The few-shot learning capabilities of sparse dictionary approaches offer potential advantages for rare disease applications, but have not been systematically evaluated in pulmonary imaging contexts.

Chapter 3

Methodology

The research consists of main phases
Overall methodology Diagram

3.1 Literature Review

following topics are covered in literature review existing solutions and their gaps research frontiers

3.2 Data acquisition and pre-processing

We selected two publicly available chest CT datasets, CT-RATE[citation] and ReXGroundingCT[citation], for training and evaluation. due to their complementary nature. CT-RATE is an open-source dataset of 50,188 reconstructed 3D chest CT volumes with their corresponding radiology reports. Each volume is annotated with 18 abnormalities, supporting extensive validation and fine-tuning. ReXGroundingCT extends CT RATE by providing expert annotated 3D segmentation masks that highlight the exact image regions referenced in the radiology reports. The dataset contains 3,142 chest CT scans across 14 abnormality categories. Both datasets are released under MIT licenses that allow their use for non-commercial academic research.

Data pre-procesising corrections such as intensity normalization and spacing adjustments have already been implemented directly in the provided NIfTI files. If further pre-processing is required, we will apply standard techniques such as resampling to a uniform voxel size, intensity normalization, and lung segmentation, before feeding the data into our model.

3.3 Model Training

Model training will be implemented using a two-phase frozen dictionary learning approach to capture both normal and pathological patterns in chest CT scans. The dataset will be divided into train,test, and validation sets.

Phase 1: Learning the Normal Dictionary We will first train a dictionary using only healthy or normal samples from the dataset. This step will allow the system to learn the typical structures of the lungs

Phase 2: Learning Pathological Atoms After the normal dictionary is established and fixed, we will train additional dictionary atoms to capture disease-specific or pathological features While the normal dictionary remains frozen, it will contribute to the sparse coding process, allowing the model to efficiently learn only the new pathological features.

We extend the dictionary learning approach using LC-KSVD to improve the discriminative power of the learned sparse representations. The label consistency constraint will produce similar sparse codes for CT samples belonging to the same disease category.

We will formulate LC-KSVD to jointly optimize three key objectives:

1. Accurate reconstruction of CT image patches using the learned dictionary.
2. Consistency of sparse representations within the same disease class.
3. Effective classification of lung conditions based on the sparse codes.

Appropriate weighting will be applied to balance these objectives, and hyperparameters will be selected through validation experiments to ensure stable and meaningful learning.

3.4 Classification and Disease Diagnosis

Support Vector Machine (SVM) use as the final disease classification model because of several features: 1. Ability to handle high-dimensional sparse features produced by dictionary learning and LC-KSVD 2. Effectiveness when the number of features is much larger than the number of samples 3. Strong generalization due to the maximum-margin principle

SVMs are also known to be robust when dealing with limited and imbalanced medical data, making them a suitable choice for disease classification tasks in chest CT analysis.

The model will be implemented using the scikit-learn library in Python and will receive the sparse codes generated from the learned dictionary as input features. A linear kernel will be used as the initial choice due to its effectiveness with sparse representations.

The classifier will predict multiple disease categories, with the number of output classes matching the labeling structure of the CT-RATE dataset.

3.5 Interpretability Analysis and Visualization

To improve interpretability and clinical relevance, the learned dictionary atoms will be visualized and analyzed through an intuitive UI/UX design that incorporates 3D rendering techniques Each dictionary atom will be mapped back to the original CT image space, allowing us to see which anatomical regions or pathological patterns the model has learned to represent. This mapping is guided by the ReXGroundingCT annotations, which provide precise spatial localization of abnormalities within the 3D imaging volume.

3.6 Volumetric Feature Extraction and Quantification

Ground-glass opacities (GGO) will be identified based on their characteristic intensity patterns in CT images, measured using Hounsfield Units (HU). GGO regions are defined as lung areas with attenuation values.then captures the partially aerated lung tissue associated with GGOs while excluding normal lung parenchyma and denser consolidations.

3.7 Baseline Comparisons and Benchmarking

To demonstrate the effectiveness of the proposed method, it will be compared against well established baseline approaches commonly used in current medical imaging research. These baselines are chosen to represent both traditional machine learning techniques and modern deep learning models Standard K-SVD is included as a baseline to highlight the advantage of incorporating label information through LC-KSVD. By comparing these two sparse representation methods, the contribution of label consistent dictionary learning to disease classification performance can be assessed. In addition, several 3D CNN architectures will be used as deep learning baselines, as they are considered state-of-the-art for volumetric CT analysis. These models are capable of learning complex spatial patterns across multiple slices and therefore provide a strong benchmark against which the proposed method can be evaluated. Existing implementations of these networks will be adopted to ensure reliability and consistency. All models will be trained and tested using the same dataset, although training strategies may differ due to the higher data and computational demands of CNN-based approaches.

Chapter 4

Timeline and Resource Required

4.1 Timeline

Provide a realistic timeline for completing the project. Use a Gantt chart or table.

4.2 Resource Required

State the resource required for the project and estimate the budget for the resources required.

Chapter 5

Conclusion

Summarize the key points of your proposal and reiterate the importance of the project.

References